A finite unit-distance obstruction in a cyclotomic module

Abstract

We give an explicit finite unit-distance graph with no proper four-colouring in the additive module $M_2 = \mathbb{Z}[w]+t\mathbb{Z}[w]+t^2\mathbb{Z}[w]$, where $w = e^{\pi i/21}$ and $t \in \mathbb{Q}(w)$ is a specified complex number of modulus one, represented with denominator 7. The construction has at most 3 535 305 907 vertices. Its coordinates have denominator 49 in the power basis of w, with all integer numerators bounded in absolute value by 3304. The geometric forcing input is Haugland's cited computational terminal-separation result. We combine this input with a finite quotient obstruction, exact cyclotomic words, and shared equality forests. Within a fixed family of twenty-one whole direction lines, we characterize four-colourability and certify that eighteen is the least number of retained lines that can give an obstruction. For the lifting words, a seven-pair representation reduces shortest length in the forty-two roots of unity to at most 147 explicit candidates. Exact intersections of disjoint same-direction gadget pairs give the final size bound. The graph is supplied as a finite expansion recipe with compact certificates rather than as an expanded vertex list.

1 Introduction and principal result

A finite quotient can make a colouring obstruction inexpensive to detect while leaving its geometric realization costly. Identifying two points modulo an additive subgroup does not require them to have the same colour in a unit-distance graph. To transfer a quotient obstruction back to the plane, one must enforce the particular identifications used by its edges. This paper carries out that transfer in a fixed cyclotomic module, using finite geometric gadgets throughout.

Let $w = e^{\pi i/21}$, let $\mathcal{O} = \mathbb{Z}[w]$, and put

$$N(X) = 2 + X + 3X^2 - 2X^4 + X^5 + 5X^6 + X^7 + X^8 - 6X^9 - 4X^{10} + 4X^{11}, \tag{1}$$

$$t = N(w)/7, \qquad M_2 = \mathcal{O} + t\mathcal{O} + t^2\mathcal{O}. \tag{2}$$

Here and below the embedding in the complex plane is the one specified by w. A unit-distance graph means a simple graph on distinct complex points whose chosen edges all have Euclidean length one. We distinguish this chosen edge set from the full induced unit-distance graph, which also includes every other unit pair among the same points.

Theorem 1.1. *There is an explicitly specified finite unit-distance graph Γ with vertex set in M_2, with no proper four-colouring, and with*

$$|V(\Gamma)| \le 3\,535\,305\,907,$$
$$|E(\Gamma)| \le 20\,579\,719\,785.$$

Every vertex has the form

$$\frac{c_0 + c_1 w + \cdots + c_{11} w^{11}}{49}, \qquad c_i \in \mathbb{Z}, \qquad |c_i| \le 3304.$$

The graph is defined by the deterministic finite recipe in Sections 4–6. Its forcing input is the terminal-separation result for Haugland's graph G_1 [3, Section 3].

The theorem includes a size bound for the selected edges of Γ. Adding all other unit pairs preserves non-four-colourability and the vertex bound, but the displayed edge bound does not count those extra pairs. We do not give a five-colouring of Γ. The construction and its proof require only the conclusion that four colours are impossible.

There are three distinct finite reductions. First, the quotient $M_2/2M_2$ has 4096 elements. Its relevant directions form twenty-one lines for a common $\mathbb{F}_4$ scalar action. A character-sum criterion characterizes exactly which selections of whole lines yield a graph without a four-colouring. A small complete deletion certificate proves that eighteen is the least possible number of retained lines in this family. We use one such choice, with 54 directions and 110 592 edges.

Second, a local equality gadget turns a permitted unit step u into the forced equality of colours at x and $x + 2u$. For each quotient edge we give an exact word whose doubled displacement connects the required representatives. A colouring of the resulting finite union would then colour the quotient. This argument uses neither a compactness theorem nor a preliminary assumption that all colourings of the ambient module are periodic.

Third, the words and their geometric realization are compressed in separate ways. The integral part of each displacement is written as a shortest word in the 42 roots w^r. Common paths are then replaced by trees on their actual algebraic coordinates. Finally, exact local intersections of disjoint same-direction gadget pairs reduce the union bound. Against the unshared power-basis word recipe for the same 54 directions, the vertex bound decreases from 5 867 216 896 to the bound in Theorem 1.1, a reduction of approximately 39.7%.

The root-of-unity word metric belongs to the established framework of Beck and Hoşten [1]. Their Theorem 3(iii) covers the $42 = 2 \cdot 3 \cdot 7$ root case, while Theorem 13 establishes the total-unimodularity fact underlying the integral linear-programming optimum. We retain a direct proof of the specific seven-pair and 147-candidate formula used here. Haugland's two-copy equality construction supplies the local geometric idea [3, Section 3]; our transport and reflection formulas keep its copies in the specified module. We make no claim of priority for the specialization or of a literature size record for the resulting graph. The size comparison concerns the two explicitly described recipes in this paper.

The construction is large, but its logical and arithmetic certificates are finite and substantially smaller than an expansion. The companion supplement describes the seed correspondence, the complete line-deletion certificate, the word and forest records, the maximum eligible pair matching, and the three local intersection checks. We separate those exact calculations from the one imported forcing result. The proof below explains precisely what each calculation has to establish and how its output enters the theorem.

2 Cyclotomic coordinates and local equalities

2.1 The module and its quotient

We compute in $K = \mathbb{Q}(w) = \mathbb{Q}[X]/(\Phi_{42})$, with

$$\Phi_{42}(X) = X^{12} + X^{11} - X^9 - X^8 + X^6 - X^4 - X^3 + X + 1.$$

The ordered power basis is $1, w, \dots, w^{11}$. Thus equality is equality of the twelve rational coefficients after reduction by Φ_{42}. Conjugation sends w to w^{-1}, and exact multiplication gives

$$t\bar{t} = 1, \qquad \rho = w^7, \qquad \rho^2 - \rho + 1 = 0, \qquad \delta = 2\rho - 1, \qquad \overline{\delta} = -\delta. \tag{3}$$

In particular, $|t| = |\rho| = 1$ and $|\delta|^2 = 3$. These identities can be checked directly from (1) in the quotient ring. Set $M_1 = \mathcal{O} + t\mathcal{O}$. Since $7t \in \mathcal{O}$ and $49t^2 \in \mathcal{O}$, we have

$$\mathcal{O} \subset M_2 \subset \frac{1}{49}\mathcal{O}. \tag{4}$$

Translations by elements of M_2 and multiplication by any power of w preserve M_2.

Lemma 2.1. *Inclusion induces an isomorphism of additive groups*

$$\mathcal{O}/2\mathcal{O} \;\simeq\; M_2/2M_2.$$

Consequently this quotient can be identified with $V = \mathbb{F}_2[X]/(\Phi_{42})$, a twelve-dimensional $\mathbb{F}_2$ vector space.

Proof. For $y \in M_2$, the element $49y$ belongs to $\mathcal{O}$ and $y - 49y = -48y \in 2M_2$, proving surjectivity. If $o \in \mathcal{O}$ equals $2y$ for $y \in M_2$, write $y = \sum_i c_i w^i/49$ with $c_i \in \mathbb{Z}$. Each $2c_i$ is divisible by 49, so each c_i is divisible by 49. Thus $y \in \mathcal{O}$ and $o \in 2\mathcal{O}$, proving injectivity. The description of $\mathcal{O}$ in its power basis gives the stated quotient. □

We denote the class of $z \in M_2$ in V by $[z]$, reserving an overbar for complex conjugation. The scalar $[\rho]$ satisfies $[\rho]^2 + [\rho] + 1 = 0$. It therefore defines an $\mathbb{F}_4$ scalar action on V. No assertion that the ring V is a field is needed. We encode $a = \sum_{i=0}^{11} a_i[X]^i$, with $a_i \in \{0, 1\}$, by the integer $\sum_i a_i 2^i$, and fix the section

$$s(a) = \sum_{i=0}^{11} a_i w^i. \tag{5}$$

2.2 The cited seed and its transport

Our seed is Haugland's graph G_1 from [3, Section 3]. In the coordinates used here it consists of 740 points of M_1, with terminals 0 and δ. Its full unit-distance edge set has 3985 edges. We use the following property reported in Haugland's preprint:

$$\text{every proper four-colouring of } G_1 \text{ assigns different colours to 0 and } \delta. \tag{6}$$

This is an imported computational forcing result. The retained coordinate and edge checks establish that our copy is the graph to which that result applies; they do not replace its colouring proof. In particular, checking all $\binom{740}{2} = 273\,430$ exact distances establishes the complete unit-edge list, whereas (6) is justified by the cited source. The supplement describes this source correspondence.

Let

$$\mathcal{U} = \{t^j w^r : 0 \le j \le 2,\ 0 \le r < 42\}.$$

For an indexed direction $u = t^j w^r$ define

$$T_u(g) = \begin{cases} ug, & j = 0, 1, \\ -u\overline{g}, & j = 2. \end{cases} \tag{7}$$

These maps are plane isometries fixing 0. By (3), each sends δ to $u\delta$. The reflection in the third shell is essential to the following containment:

$$\begin{aligned} M_1 \subset M_2, \qquad tM_1 = t\mathcal{O} + t^2\mathcal{O} \subset M_2, \\ t^2\overline{M_1} = t^2\mathcal{O} + t\mathcal{O} \subset M_2. \end{aligned}$$

Here $\overline{t} = t^{-1}$ and $\overline{\mathcal{O}} = \mathcal{O}$. Hence $x + T_u(G_1)$ lies in M_2 for every $x \in M_2$. Restricting any four-colouring to this copy yields different colours at x and $x + \delta u$.

2.3 A finite equality gadget

For $x \in M_2$ and $u \in \mathcal{U}$, set

$$A = x, \quad B = x + 2u, \quad C = x + u, \quad D = x + \rho u, \quad E = x + (1 + \rho)u. \tag{8}$$

Define $H(x, u)$ to contain these points, the seven unit edges

$$CD,\ CE,\ DE,\ AC,\ AD,\ BC,\ BE, \tag{9}$$

and the two full seed copies

$$A + T_{\rho^{-1}u}(G_1), \qquad D + T_{-\rho u}(G_1). \tag{10}$$

Coincident coordinates are identified, and repeated edges are retained once. The parameters $\rho^{-1}u$ and $-\rho u$ are still indexed directions in the same shell. All selected edges have length one: the seed copies are isometric, and the differences in (9) are, up to sign, u, ρu, or $(1-\rho)u$. This is the two-copy equality arrangement from [3, Section 3], transported as in (7).

Lemma 2.2. *Every proper four-colouring of $H(x, u)$ gives the same colour to x and $x + 2u$. All vertices of $H(x, u)$ belong to M_2.*

Proof. The first copy in (10) has terminals A, E, because $\delta\rho^{-1} = 1 + \rho$. The second has terminals D, B, because $-\delta\rho = 2 - \rho$. The cited seed property therefore gives $c(A) \neq c(E)$ and $c(D) \neq c(B)$. The triangle CDE uses three distinct colours. The four real edges incident with A or B in (9), together with the two seed inequalities, make each of A, B avoid all three colours. Both use the fourth colour. Membership follows from the transport containment and the five explicit points in (8). □

There is a useful logical distinction here. The virtual constraint graph on A, B, C, D, E, obtained by adding the two inequalities AE and DB, is K_5 with AB removed. Its two-terminal relation is exactly $\mathrm{EQ} = \{(c, c) : c \in \{1, 2, 3, 4\}\}$. For the actual geometric graph $H(x, u)$, Lemma 2.2 proves only

$$R_4(H(x, u), \{A, B\}) \subseteq \mathrm{EQ}\,. \tag{11}$$

The left side denotes boundary assignments that extend to a full proper colouring; it may be empty. The seed interiors have physical intersections, and compatible extension of a virtual colouring has not been established. Every subsequent use of the gadget is in the necessary direction (11).

3 The quotient and the eighteen-line threshold

3.1 A character-sum characterization

The 63 residues

$$[t^j w^r], \qquad 0 \leq j \leq 2, \quad 0 \leq r < 21,$$

are distinct and nonzero. They split into 21 disjoint sets $L = \{v, [\rho]v, [\rho]^2 v\}$, the nonzero vectors on $\mathbb{F}_4$ lines. Write $\mathcal{L}$ for this fixed family. These assertions are exact finite arithmetic checks from (1); the supplement supplies the ordered triples as twelve-bit integers. Given $S \subseteq \mathcal{L}$, define

$$D_S = \bigcup_{L \in S} L, \qquad Q_S = \mathrm{Cay}(V, D_S).$$

Thus a, b are adjacent when $a + b \in D_S$. Since V has characteristic two and $0 \notin D_S$, this is a simple undirected $3|S|$-regular graph on 4096 vertices.

For a nonzero binary linear functional $a \in V^*$ put

$$H_a = \{L \in \mathcal{L} : a(v) = 0 \text{ for every } v \in L\}. \tag{12}$$

Fix S, and write $m = |S|$ and $b_a = |S \cap H_a|$. The additive character $\chi_a(v) = (-1)^{a(v)}$ is an adjacency eigenvector of Q_S. Its sum over a whole line equals 3 if a annihilates the line and equals -1 otherwise: a nonzero functional on a two-dimensional binary space vanishes at exactly one of its three nonzero vectors. It follows that

$$\lambda_a = 4b_a - m \quad (a \neq 0), \qquad \lambda_0 = 3m. \tag{13}$$

This computation uses the common scalar action, and gives every eigenvalue because the 4096 additive characters form a basis.

Theorem 3.1. *For $S \subseteq \mathcal{L}$, the graph Q_S has no proper four-colouring if and only if*

$$S \cap H_a \neq \varnothing \qquad \text{for every } a \neq 0. \tag{14}$$

If (14) *fails, Q_S has an additive four-colouring.*

Proof. Suppose first that a nonzero h has $S \cap H_h = \varnothing$. Colour $v \in V$ by the pair

$$(h(v), h([\rho]v)) \in \mathbb{F}_2^2. \tag{15}$$

If a direction $d \in D_S$ had colour $(0, 0)$, then h would vanish on d and $[\rho]d$, and also on $[\rho]^2 d = d + [\rho]d$. It would annihilate the selected whole line containing d, a contradiction. The map in (15) is additive, so the colours at the ends of every edge differ. This proves the claimed colouring, including the case $S = \varnothing$.

Now suppose (14) holds. The graph is nonempty. Its least eigenvalue λ is negative, since a nonempty regular loopless graph has a positive constant eigenvalue and the sum of its eigenvalues is zero. Formula (13) gives $\lambda \geq 4 - m$. In particular $m > 4$. Write $q = -\lambda$, so $0 < q \leq m - 4 < m$. Hoffman's ratio bound, in the form stated in [2, Theorem 1], gives

$$\alpha(Q_S) \leq 4096 \frac{q}{3m + q} < \frac{4096}{4}.$$

All its hypotheses hold: Q_S is finite, simple, regular, nonempty, and has negative least eigenvalue. Four independent colour classes cannot cover its vertices. □

The converse direction is more specific than the spectral bound. For an arbitrary family of binary two-dimensional subspaces, the character sums still have the same form, but there need not be one common scalar $[\rho]$ that supplies (15). That common action is what turns the spectral sufficient condition into the exact characterization needed for the finite classification.

3.2 A complete deletion certificate

Let $D = \mathcal{L} \setminus S$ be the set of deleted lines. Condition (14) is equivalent to

$$H_a \nsubseteq D \qquad (a \neq 0). \tag{16}$$

It suffices to keep the inclusion-minimal distinct sets among the H_a. Let $\mathcal{H}$ be the resulting hypergraph on the 21 lines. The admissible deletion sets are exactly its independent sets. The following elementary criterion makes a supplied complete list checkable without appealing to a search program's assertion that it found everything.

Lemma 3.2. *Let $\mathcal{H}$ be a hypergraph on a finite linearly ordered set. Suppose a family $\mathcal{I}$ contains the empty set, every member is independent, and whenever $I \in \mathcal{I}$ and $x > \max I$ with $I \cup \{x\}$ independent, the set $I \cup \{x\}$ also belongs to $\mathcal{I}$. For $I = \varnothing$ allow every x. Then $\mathcal{I}$ is the complete family of independent sets of $\mathcal{H}$.*

Proof. If $J = \{x_1 < \cdots < x_r\}$ is independent, then so is every prefix $\{x_1, \ldots, x_i\}$. Starting at the empty set and applying the extension condition proves inductively that each prefix, and finally J, belongs to $\mathcal{I}$. □

Proposition 3.3. *For the fixed family $\mathcal{L}$, exactly eighteen retained whole lines are needed at minimum to obtain a graph without a proper four-colouring. The numbers of non-four-colourable selections of* $18, 19, 20, 21$ *lines are respectively* $210, 147, 21, 1$*. Every other selection is four-colourable.*

Proof. Exact evaluation of the 4095 nonzero functionals in (12) gives 1288 distinct masks, all nonempty, with minimum cardinality two. Their inclusion-minimal subfamily consists of 63 pairs and 133 triples. The accompanying independent-family certificate has the rank counts in Table 1. Every listed set avoids the 196 forbidden masks, and every allowed increasing extension is present. There are 378 such extensions. The empty set is present, so Lemma 3.2 proves completeness. In particular there are no admissible deletion sets of size four or greater. Complements and Theorem 3.1 give the result. □

Number of deleted lines	0	1	2	3
Number of admissible sets	1	21	147	210

Table 1: The complete 379-member independent family for the forbidden deletion hypergraph. The data and increasing-extension checks are supplied in the companion certificate.

The computation in this proposition has two separate obligations. The forbidden masks must be derived from all nonzero functionals and the specified line family; and the listed independent sets must be complete for those masks. The certificate and its checker perform both operations. It would be insufficient to verify only that a proposed set of eighteen lines meets every mask, since that would give an example without the minimality statement. Conversely, the classification concerns whole lines in $\mathcal{L}$; it is not a minimality statement for arbitrary quotient edges, vertices, or finite geometric obstructions.

3.3 The selected eighteen-line graph

Order the triples of $\mathcal{L}$ lexicographically by their sorted integer encodings. For the geometric construction we delete the three triples

$$\{32, 2907, 2939\}, \qquad \{972, 2154, 2982\}, \qquad \{1398, 2236, 3530\}. \tag{17}$$

They have zero-based indices $5, 15, 17$. Write S_* for the remaining eighteen lines and $Q = Q_{S_*}$. The graph has 54 directions and $4096 \cdot 54/2 = 110\,592$ edges. Direct character sums give Table 2; in particular the least eigenvalue is -14. Thus

$$\alpha(Q) \leq 4096 \frac{14}{54 + 14} = \frac{14336}{17} < 1024. \tag{18}$$

The spectrum is for the particular deletion (17). Other eighteen-line choices can have different multiplicities despite having the same degree and least eigenvalue.

Eigenvalue	-14	-10	-6	-2	2	6	10	14	18	54
Multiplicity	135	477	693	864	693	558	459	180	36	1

Table 2: The spectrum of the selected quotient Q.

The deletion in (17) minimizes the original power-basis period cost among the 379 admissible deletion masks. This explains its selection, but no optimality of that choice is needed for the geometric contradiction. Formula (18) independently certifies the exact graph we use. Nor does the selection limit the final unit-distance graph to 54 geometric directions: the seed copies and equality paths retain their own unit edges.

4 Shortest words in the forty-two roots

The length of $n \in \mathcal{O}$ in the generating set $\{w^r : 0 \le r < 42\}$ is the least number of these roots whose sum is n. All roots are allowed with nonnegative multiplicities; their negatives are already in the generating set. Beck and Hoşten study precisely this root-of-unity word metric [1]. For $42 = 2 \cdot 3 \cdot 7$, their total-unimodularity result implies that the multiplicity problem

$$\min\{\mathbf{1}^T x : Ax = b,\ x \ge 0,\ x \in \mathbb{Z}^{42}\} \tag{19}$$

has the same value as its real relaxation, with A expressed in their integral coordinates. Indeed a feasible word exists, and intersecting the feasible region with a cost sublevel set gives a compact nonempty set. An optimal basic feasible solution is integral by the determinant criterion. See [1, Theorems 3(iii) and 13].

Changing the integral basis preserves the feasible multiplicity vectors and their costs; it does not justify asserting that every changed generator matrix is itself totally unimodular. Also, an arbitrary fractional point on the optimal face is not an integral basic optimum. For the fixed instance here it is convenient to avoid selecting LP bases altogether. We now derive the explicit finite formula used by the word certificate.

4.1 Seven Eisenstein pairs

Put $\zeta = w^6$, of order seven. Since $\rho = w^7$ has order six and $w = \rho\zeta^{-1}$, one has $K = \mathbb{Q}(\zeta, \rho)$. The degree of K over $\mathbb{Q}$ is twelve and that of $\mathbb{Q}(\rho)$ is two. Thus the minimal polynomial of ζ over $\mathbb{Q}(\rho)$ is $\Phi_7(X) = 1 + X + \cdots + X^6$.

Every $n \in \mathcal{O}$ admits a representation

$$n = \sum_{k=0}^{6} (a_k + b_k\rho)\zeta^k, \qquad a_k, b_k \in \mathbb{Z}. \tag{20}$$

For example, for each power w^r choose $h \equiv r \pmod 6$, $0 \le h < 6$, and $k \equiv (r - 7h)/6 \pmod 7$. Then $w^r = \rho^h\zeta^k$, and the six powers of ρ are

$$1, \quad \rho, \quad \rho - 1, \quad -1, \quad -\rho, \quad 1 - \rho. \tag{21}$$

Applying this rule to the twelve power-basis coefficients fixes a deterministic initial representation.

Lemma 4.1. *Two integral representations* (20) *of the same n differ by a common pair shift: for some $A, B \in \mathbb{Z}$, their pairs are (a_k, b_k) and $(a_k + A, b_k + B)$ for every k.*

Proof. The difference is a polynomial of degree at most six over $\mathbb{Q}(\rho)$ that vanishes at ζ. It is therefore $C\Phi_7$ for some $C \in \mathbb{Q}(\rho)$. All seven coefficients of this polynomial lie in $\mathbb{Z}[\rho]$, so $C = A + B\rho$ with $A, B \in \mathbb{Z}$. Conversely adding a common pair changes the sum by $(A + B\rho)\Phi_7(\zeta) = 0$. □

Lemma 4.2. *The least number of letters from* $\{\pm 1, \pm\rho, \pm(\rho - 1)\}$ *summing to* $a + b\rho$ *is*

$$h(a, b) = \max\{|a|, |b|, |a + b|\} = \frac{|a| + |b| + |a + b|}{2}. \tag{22}$$

Proof. Each permitted letter changes each of the coordinates $a, b, a + b$ by at most one. This gives the lower bound by the displayed maximum. If a, b have the same sign, use $|a|$ copies of the corresponding signed 1 and $|b|$ copies of the corresponding signed ρ. If their signs are opposite, use $\min(|a|, |b|)$ copies of the appropriate signed $(\rho - 1)$; the remaining displacement is along one signed basis direction. These words attain the maximum in both cases. The identity with the half-sum follows from the same two cases. □

The coprime orders seven and six partition the 42 roots into the seven classes ζ^k times (21). A word can therefore be grouped into seven local displacements. Lemma 4.1 describes all possible resulting pair representations, including their integral kernel, rather than only their rational dimension.

Theorem 4.3. *For an integral representation* (20)*, the shortest root-word length of* n *is*

$$\ell(n) = \min_{(A,B)\in\mathbb{Z}^2} f(A, B), \qquad f(A, B) = \sum_{k=0}^{6} h(a_k + A, b_k + B). \tag{23}$$

The minimum is attained among the intersections of nonparallel lines in

$$A = -a_k, \qquad B = -b_k, \qquad A + B = -a_k - b_k \quad (0 \le k \le 6). \tag{24}$$

There are at most 147 *candidate intersections, all integral.*

Proof. Grouping an arbitrary word into its seven root classes and using Lemmas 4.1 and 4.2 shows that its length is at least $f(A, B)$ for some integral shift. Conversely, for any shift take a shortest six-direction word in each class and concatenate them. The sum is n, and the length is $f(A, B)$. This proves (23).

Regard f on $\mathbb{R}^2$. By the half-sum formula in (22), it is continuous, convex, and affine on each closed cell of the line arrangement (24). It is coercive: the summands bound $|A| - \max_k |a_k|$ and $|B| - \max_k |b_k|$ below, so $f(A, B)$ tends to infinity whenever $\max(|A|, |B|)$ does. Hence it attains its real minimum, and its minimum-level set is bounded.

Choose a closed arrangement cell containing a minimizer. On that cell f is affine, and the intersection with the minimum-level set is a nonempty bounded face. Such a face has an extreme point, which is a vertex of the cell. A cell vertex is the intersection of two nonparallel arrangement lines. This argument includes the possibility that the minimum is attained along an edge or throughout a polygon. Every such intersection is integral. There are at most $7 \cdot 7$ vertical–horizontal pairs, $7 \cdot 7$ vertical–diagonal pairs, and $7 \cdot 7$ horizontal–diagonal pairs. Thus the real minimum occurs at one of at most 147 integral candidates and equals the integral minimum in (23). □

4.2 Deterministic counts and their cost

The retained generator chooses the lexicographically first minimum (A, B) among the candidate points. For each shifted pair, use the word described in Lemma 4.2: cancel opposite signs first using $\rho - 1$ or $1 - \rho$, then use the remaining signed 1 and ρ letters. Combining all seven classes gives 42 nonnegative multiplicities c_r, with

$$n = \sum_{r=0}^{41} c_r w^r, \qquad \sum_{r=0}^{41} c_r = \ell(n). \tag{25}$$

At this stage only multiplicities are fixed; the order in which the letters are traversed will be chosen in the finite construction. For the particular certificate expansion in this paper, the stored multiplicities in `ROOT_WORDS.json` are the defining inputs. The rule above describes their generation. The independent replay checks each stored word identity and minimum length; it does not separately check the canonical tie-break or the recorded candidate count. These extra generation choices are unnecessary for the lifting proof.

The input comprises twelve integer coefficients. If their bit length is at most B, all pairs, candidate shifts, and output counts have $O(B)$ bits. The number of candidates and arithmetic operations is fixed, independently of the input magnitudes. Thus the compressed count output has polynomial bit cost, using exact additions and comparisons on these integers. This is a statement about multiplicities, not about writing every letter. For a positive integer n, the triangle inequality shows $\ell(n) \geq n$, and the word of n copies of 1 attains equality. Explicit output length can therefore be exponential in the bit length of the input.

The optimization is also deliberately restricted. It solves the integral root-word subproblem exactly. It does not optimize an arbitrary word in all of $\mathcal{U}$, the nonintegral tail below, the section s, the quotient selection, or the final geometric union.

5 Finite lifting and shared equality forests

5.1 Words for quotient edges

Each of the 54 retained quotient directions has a unique indexed lift $u = t^j w^r$ with $0 \leq j \leq 2$ and $0 \leq r < 21$. For each quotient edge, write $b = a + [u]$ and orient it by the integer encoding so that $a < b$. Let

$$d = 7^j, \qquad U = du \in \mathcal{O}, \qquad k = (d-1)/2, \qquad n = \frac{s(a) - s(b) + U}{2}. \tag{26}$$

The coefficients of n are integral: d is odd, so $[U] = [u]$, and $a + b + [u] = 0$. Direct rearrangement gives

$$\frac{s(a) + u - s(b)}{2} = n - ku. \tag{27}$$

Take any root word for n and append k copies of $-u$. Starting at $s(b)$ and doubling every word step gives a path ending at

$$p_{ab} = s(a) + u. \tag{28}$$

Every step is $2v$ for $v \in \mathcal{U}$. Placing $H(x, v)$ at such a step forces equal colours at its ends. Finally, the actual unit edge $\{s(a), p_{ab}\}$ enforces the desired quotient inequality once p_{ab} and $s(b)$ have the same colour. A zero-length word needs no equality gadget.

This is a finite construction for each of the 110 592 quotient edges. Before sharing, it gives a union of finite paths of equality gadgets. All translations lie in M_2, and Lemma 2.2 keeps their interiors there as well. The remaining issue is to share these paths without making unsupported assumptions about colouring extensions across overlapping interiors.

5.2 Physical interfaces and the direction of implication

For a finite graph G and a boundary $B \subseteq V(G)$, let $R_4(G, B)$ be the set of assignments $B \to \{1, 2, 3, 4\}$ that extend to a proper colouring of all of G. A boundary records the simultaneous colours of all its vertices. The next elementary gluing rule states the information needed for an exact relation, even though the construction will use a weaker implication.

Lemma 5.1. *Let G be the union of finite pieces G_i on physical vertex sets V_i, together with an extra edge set E_*. Coincident coordinates are first identified. Let B be the desired output boundary. Choose $B_i \subseteq V_i$ to contain $V_i \cap B$, every vertex shared with another piece, and every vertex of V_i incident to an edge of E_*. Then*

$$R_4(G, B) = \mathrm{pr}_B \left(\bowtie_i R_4(G_i, B_i) \;\; \bowtie! \bowtie_{pq \in E_*} \{(c_p, c_q) : c_p \neq c_q\} \right). \tag{29}$$

Here a natural join requires equal values at every common physical vertex, and projection existentially forgets the remaining colours.

Proof. A colouring of G restricts to every factor on the right. Conversely, take a joined assignment and choose a witness extension in each piece. All vertices common to two pieces belong to both corresponding boundaries. The witness colourings therefore agree on every physical intersection and unite into one function. This function is proper on each piece edge and, by the added inequality factors, on every extra edge. Projecting to B proves the equality. □

One witness per complete boundary assignment is enough for this proof. Keeping only smaller projections can lose the compatibility conditions, and forgetting a vertex before its last occurrence or unprocessed incident edge invalidates the gluing argument. Likewise, coincident occurrence names represent one colour variable, not two independent ones. If the full induced unit graph is intended, unit edges missing from the chosen piece lists must enter as factors in E_*.

Palette symmetry allows a relation to be encoded by equality partitions of the complete boundary into at most four blocks. At a join, one must restrict a common global partition to the local boundaries. Independently canonicalized colour names cannot simply be matched as literal labels. These observations do not provide a small relation table: on a boundary of size b a labelled table can have 4^b entries, and the complete tables for our actual gadgets are not being computed.

For non-four-colourability we need less than (29). It suffices to reason about the restriction of a single hypothetical global colouring. That is why the one-sided gadget relation (11) is adequate.

Lemma 5.2. *Let P be a finite graph on exact points of M_2, each of whose edges is realized by a chosen occurrence $H(x, u)$ with ends $x, x + 2u$. Let G contain the geometric union of those occurrences, with any additional unit edges. Every proper four-colouring of G is constant on each connected component of P.*

Proof. Restrict the same colouring to each gadget. Lemma 2.2 gives equal colours across its skeleton edge. Equality propagates along every finite path in P. □

The interiors of the pieces need not be disjoint. Extra identifications or edges may remove colourings, but cannot create a colouring that violates one of these necessary local equalities. In particular, replacing P by a spanning forest preserves the component implication without claiming to preserve the exact terminal relation or the entire colouring set of the old geometric graph. The chosen directed placement still matters when we count intersections; reversing an equality statement alone is not an identity of geometric gadgets.

5.3 The deterministic forest recipe

For each edge use the stored multiplicities for its target in `ROOT_WORDS.json`, satisfying (25). We consider exactly four orders of the 42 root exponents: increasing order; decreasing order; the order

$$(6k + 7h + 21\varepsilon) \bmod 42 \quad \text{for } k = 0, \ldots, 6,\ h = 0, 1, 2,\ \varepsilon = 0, 1 \tag{30}$$

in the indicated nested order; and its reverse. Each is a permutation of $0, \ldots, 41$. To form an edge word, list each root with its fixed multiplicity in the chosen order, then append the unchanged k copies of $-u$ from (26).

For a fixed $b \in V$, collect every oriented edge with larger endpoint b. Traverse its direction records in increasing (j, r) order. For each of the four root orders, count the distinct nonempty prefixes of these words, viewed as words of indexed directions. Choose the order with the fewest prefixes, breaking ties by the listed order index. This is a finite choice from four alternatives. It does not minimize the final geometric union over all possible permutations.

Now start a set of seen coordinates with $\{s(b)\}$. Walk each of the specified doubled paths from $s(b)$. At a step $x \longmapsto y = x + 2v$, retain the occurrence $H(x, v)$ if and only if y has not yet been seen; in that case add y to the seen set. If y was already seen, retain no new gadget, but continue the prescribed path from y. The state consists of exact coordinates, not merely of word prefixes.

Lemma 5.3. *For each b, the retained skeleton edges form a tree on its seen set, rooted at $s(b)$. They connect $s(b)$ to every required endpoint p_{ab} of* (28).

Proof. At the start, the only seen point is the root. Inductively the current point x is seen and connected to the root: this holds at the start of each path and after every step. If y is new, a single retained edge attaches it to the existing tree. If y is old, it is already connected, so omitting the current edge changes nothing needed by the invariant. Each path terminates at its required endpoint by (27). Thus that endpoint is seen and connected. Every retained edge adds precisely one new vertex, proving the tree claim. □

All doubled paths in the b construction have residue b modulo $2M_2$. Trees for different b therefore have disjoint skeleton coordinates. Their gadget interiors may still coincide, and the proof does not require otherwise. Prefix sharing accounts for repeated words; the seen-set operation additionally recognizes distinct words arriving at the same exact point.

Define Γ to be the coordinate union of the 4096 representatives, all retained gadget occurrences, and all edges $\{s(a), p_{ab}\}$ for edges of Q. Exact equal coordinates are identified. Each selected edge has squared length one, so none becomes a loop after correct coordinate identification.

Proposition 5.4. *The finite graph Γ has no proper four-colouring.*

Proof. If c were such a colouring, Lemmas 5.2 and 5.3 would give $c(p_{ab}) = c(s(b))$ for each edge of Q. The selected unit edge $\{s(a), p_{ab}\}$ would give $c(s(a)) \neq c(s(b))$. Hence $a \mapsto c(s(a))$ would properly four-colour Q, contradicting (18). □

This proof concerns only the finitely many paths and copies just defined. It does not extract a graph from a hypothetical infinite colouring, and it does not assemble independent colouring extensions of the pieces. Those two distinctions make the argument robust under the extensive physical sharing used to reduce the size.

6 Size, coordinates, and reproducibility

6.1 Local intersection counts

The canonical graph $H(x,u)$ has fewer vertices than the separate pieces in its definition would suggest, because some coordinates coincide. Exact local enumeration gives the following data. They refer throughout to the selected edge sets, with the complete seed edge lists included.

Proposition 6.1. *For each $j = 0, 1, 2$, the graph $H(0, t^j)$ has* 1066 *vertices and* 6071 *selected edges. The union*

$$J_j = H(0, t^j) \cup H(2t^j, t^j)$$

has 2045 *vertices and* 11944 *selected edges. Its two occurrences share* 87 *vertices and* 198 *selected edges. The same counts hold for $H(x,u)$ and $H(x,u) \cup H(x+2u,u)$ for every permitted indexed direction $u = t^j w^r$ and $x \in M_2$.*

Proof. For each of the three values of j, form the five skeleton points and the two lists of 740 transported seed coordinates from (10). Reduce coordinates to the common denominator 49, identify identical twelve-integer tuples, and deduplicate the seven skeleton edges and the transported seed edges. This gives 1066 vertices and 6071 edges in every case. Repeat for the two occurrences with anchors 0 and $2t^j$, retaining the maps from each local list into the union. The resulting coordinate intersections have cardinality 87, and the selected edge intersections have cardinality 198. Thus $2 \cdot 1066 - 87 = 2045$ and $2 \cdot 6071 - 198 = 11944$. The companion local-geometry certificate contains the coordinate and edge lists and their incidence maps; each selected edge has been checked by exact norm multiplication.

For uniformity, the definitions give the identity of vertex and selected edge sets

$$H(x, t^j w^r) = x + w^r H(0, t^j), \tag{31}$$

also in the reflected shell $j = 2$. Apply the same translation and rotation to both occurrences of J_j. This proves the remaining counts. □

Haugland reports 6264 induced unit edges for the source graph G_2 [3, Section 3]; the 6071 edges of H counted here are the selected edges specified in its definition. The edge conventions differ. The local counts are finite computational ingredients, not deductions from the equality relation alone. An abstract edge enforcing the same equality could have entirely different geometry. In the present two-copy union the full induced graph has 12697 edges, 753 more than the selected union. We use 11944 in the selected-edge count. Extra induced edges are unnecessary to the proof, and no induced-edge count for the large network is inferred from this local example.

6.2 Maximum disjoint pair packaging

The retained occurrence stream fixes the graph, but its traversal order need not constrain an upper bound for the final union. Within each fibre, make an auxiliary graph whose vertices are the retained occurrences. Join the occurrences $H(x,u)$ and $H(y,u)$ precisely when the same indexed directed unit $u = t^j w^r$ occurs in both and $y = x + 2u$ as an exact twelve-coordinate identity. This is exactly the placement covered by Proposition 6.1; a shared residue or an oppositely oriented equality is insufficient.

Each occurrence has at most one predecessor and one successor in this graph: their anchors, if present, must be $x - 2u$ and $x + 2u$, respectively. There cannot be a directed cycle, since $x + 2nu = x$ would imply $u = 0$ for some $n > 0$. Consequently every component is a path. Pairing adjacent vertices from one end of each path yields a maximum disjoint

matching of $\lfloor m/2 \rfloor$ pairs on a path of m occurrences. The exact forest replay constructs these paths over all 4096 fibres and obtains $P = 1\,460\,199$ pairs. Some paired occurrences are nonconsecutive in the original traversal stream; the graph and all its coordinates remain unchanged.

Let T be the total number of retained occurrences, and let S be the equality skeleton. Each fibre's retained edges form a tree on its exact points, and different fibres have distinct residues modulo $2M_2$; hence $|S| = 4096 + T$. Every occurrence has two distinct endpoints already in S, so it contributes at most $1066 - 2 = 1064$ further vertices. Every eligible paired union has three distinct skeleton endpoints and contributes at most $2045 - 3 = 2042$ further vertices. Partition the occurrences into P disjoint pairs and $T - 2P$ singletons. Applying the union bound to the vertices outside S, and separately to selected edges, gives

$$|V(\Gamma)| \le (4096 + T) + 1064(T - 2P) + 2042P = 4096 + 1065T - 86P, \tag{32}$$

$$|E(\Gamma)| \le 110592 + 6071(T - 2P) + 11944P = 110592 + 6071T - 198P. \tag{33}$$

The factor 86 is one less than the 87 shared vertices because the three skeleton points of a pair are counted in S. Further coincidences between different blocks can only improve these bounds. Disjointness here concerns occurrences, not their physical vertex sets; the edge count is for selected edges only.

These estimates also explain why pair orientation is checked. Equality of colours is symmetric, but the placement $H(x, u)$ is a directed geometric construction. Only the actual same-direction pair in Proposition 6.1, or its simultaneous translate and rotation, justifies subtracting the stated intersection counts.

6.3 The exact forest ledger

The specified quotient and section produce 2835 distinct integral targets n in (26). For each target the word record stores the selected shift, the 42 multiplicities, and the number of distinct candidates tested. The independent replay checks the identity (25) and minimum length by a separate integer-search oracle. The forest records then specify one of the four orders for every $b \in V$ and give the per-fibre counts. Reconstructing the doubled paths checks every one of the 110 592 endpoint identities.

The resulting totals are displayed in Table 3. A root-word occurrence is counted with multiplicity before any sharing. The prefix total counts nonempty prefixes in the selected per-fibre tries. The forest total counts only steps whose endpoint is a new exact coordinate. These are different quantities and need not agree.

Quantity	Value
Quotient edges and checked lift identities	110 592
Distinct integral targets	2 835
Shortened full-word occurrences before sharing	3 649 656
Shared prefix occurrences	3 439 213
Retained exact-coordinate forest edges T	3 437 445
Maximum disjoint eligible same-direction pairs P	1 460 199
Maximum shortest integral root-word length	66
Maximum shortened full-word length, including tail	90
Maximum absolute anchor or endpoint numerator over 49	3 087

Table 3: Exact totals for the fixed deterministic construction. The word and forest records retain the per-target and per-fibre data.

Substituting the certified values of T and P in (32)–(33) gives

$$4096 + 1065 \cdot 3\,437\,445 - 86 \cdot 1\,460\,199 = 3\,535\,305\,907,$$
$$110592 + 6071 \cdot 3\,437\,445 - 198 \cdot 1\,460\,199 = 20\,579\,719\,785.$$

Without the pair intersections, the same forest would give the vertex bound $3\,660\,883\,021$. Thus shortening, coordinate sharing, and local block counting contribute separately to the result.

For comparison, keep the same quotient and write $n = \sum_{i=0}^{11} n_i w^i$ using $|n_i|$ signed basis letters, followed by the same tail. With independent equality paths this recipe uses $5\,509\,120$ occurrences. The local count of 1066 vertices per gadget then gives

$$4096 + 1065 \cdot 5\,509\,120 = 5\,867\,216\,896.$$

The maximum full-word length in this comparison is 198, whereas the shortened recipe has maximum 90. Both numbers use the same 54 quotient directions. The optimization statement of Theorem 4.3 concerns each integral root target; none of these global counts asserts optimality of the entire forest or graph.

6.4 A uniform coordinate bound

For $z \in \frac{1}{49}\mathcal{O}$, write $z = (c_0 + c_1 w + \cdots + c_{11} w^{11})/49$ and define

$$\mathrm{ht}_{49}(z) = \max_i |c_i|.$$

Exact enumeration of the finite unit and seed transform lists gives

$$\max_{u \in \mathcal{U}} \mathrm{ht}_{49}(u) \le 63, \qquad \max_{u \in \mathcal{U},\ g \in V(G_1)} \mathrm{ht}_{49}(T_u(g)) \le 154. \tag{34}$$

This includes the reflected shell and checks integrality at denominator 49. In $H(x, u)$ a skeleton offset from x is a sum of at most two generating units. A transported seed offset is either a seed image or one generating unit plus a seed image. Therefore every local vertex satisfies

$$\mathrm{ht}_{49}(z - x) \le \max(2 \cdot 63, 63 + 154) = 217. \tag{35}$$

For the shortened paths, coordinatewise monotonicity of the old basis word is no longer available. Instead, the forest check records the exact maximum of all retained anchors and endpoints, which is 3087. Every point ever visited in a fibre is either its representative or was first inserted as a retained endpoint. Representatives have height at most 49. This proves that the bound 3087 covers all visited points, including those reached later by omitted steps, and in particular every required p_{ab}. Combining it with (35) gives

$$\mathrm{ht}_{49}(z) \le 3087 + 217 = 3304 \qquad (z \in V(\Gamma)).$$

Together with Proposition 5.4 and the counting formulas, this completes the proof of Theorem 1.1.

The coordinate bound does not grow with the number of occurrences. Each numerator fits in thirteen signed bits; the large size is the number of geometric pieces, rather than a need for high-precision coordinates. These observations use a fixed algebraic basis. They make no assumption of unit-cost arithmetic on arbitrary real numbers. For a difference represented over 49, the unit test is the exact identity that its numerator times its conjugate equals 2401 in the cyclotomic quotient.

6.5 What the certificates represent

The companion certificate archive contains the seed coordinates and complete edge list, the fixed line data and complete deletion family, the chosen quotient, the local pair geometry, the word records, the forest configuration and counts, and the arithmetic and generator code. The supplement gives their correspondence to the assertions in this paper. In particular, `ROOT_WORDS.json` records the 2835 targets, and `SHORT_FOREST.json` records the choices and counts for all 4096 fibres. These two files alone are not the complete dependency set: the seed, exact arithmetic, and local geometry are also needed.

To expand Γ, initialize the representatives, iterate the retained forest occurrences, emit each canonical $H(x, u)$, identify identical coordinate tuples, and add the 110 592 lifted unit edges. The procedure terminates because every loop has the finite bounds given above. A dictionary or external sorting can identify repeated tuples. There is no need to test every pair of final vertices: the selected edges already establish the obstruction. Expansion nevertheless scales with millions of local occurrences and potentially billions of vertices; it has not been carried out. A digest of the deterministic occurrence stream binds the compact records to an intended stream, but is not a substitute for its generator or for the mathematical proof.

There are consequently three forms of evidence with different roles. The cited seed result supplies the terminal inequality. Exact local arithmetic supplies source correspondence, unit distances, intersections, the quotient spectrum, word identities, and the finite counting ledger. The proofs in this paper show how those facts force the global contradiction and the stated bounds. The retained data are not a fresh SAT refutation for the seed or the unexpanded graph.

The exact optimization statements remain within their declared finite domains: eighteen whole lines in the specified family, shortest length for an integral target in the 42 roots, and the largest disjoint matching of eligible same-indexed-direction two-gadget pairs in this frozen forest. The last statement does not optimize larger blocks, different overlap types, word orders or the graph's actual size. Further changes to the section, the fixed odd-denominator tail, or the quotient family would require their own analysis. The present result is the finite module-specific construction and its checked quantitative bounds.

References

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